

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO5_AT2 — VISUALIZATION ANALYSIS REPORT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO5 – Evaluation (BL5)	Assessment:	Assessment Tool 2 – Visualization Analysis Report
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO5 Statement:	<i>Evaluate the effectiveness of visualization design choices — proportional ink, axis scales, overplotting solutions, color use, and accessibility — and justify improvements.(BL5)</i>		
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Section / Batch:	SLOT C	Date:	27/08/2026

Code of Conduct

I Jothika M (192324235) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *Jothika M*

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond describing the scenario: explain WHY the design choice is flawed, and justify your judgment.
- Where a question asks for a fix, propose a specific improvement — not just “make it better.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – VISUALIZATION ANALYSIS REPORT

Problem 31 (5 Marks)

Data: Earthquake magnitudes converted to energy release: 8, 250, 6000, 180000.

Task: Calculate the ratio between each consecutive value, and explain why a logarithmic axis is necessary here.

Problem 32 (5 Marks)

Data: Survey results: Passed = 76%, Failed = 15%, Absent = 9%.

Task: Describe how you would build a 10×10 waffle chart for this data, stating exactly how many cells each category should fill.

Problem 33 (5 Marks)

Data: A scatterplot of 20,000 stock trades plotted as price vs. volume appears as a single solid blob.

Task: Suggest two different techniques to fix this overplotting problem, and briefly explain how each works.

Problem 34 (5 Marks)

Data: A categorical scatterplot of satisfaction ratings across 5 store branches shows heavy point-stacking directly above each category label.

Task: Explain how combining jittering with partial transparency would improve this chart's readability.

Problem 35 (5 Marks)

Data: A dataset of 9,000 (temperature, ice cream sales) point pairs is too dense to read as a raw scatterplot.

Task: Explain what each cell's color would represent in a 2D histogram of this data, and what a very dark cell would indicate.

Rubrics for CalculationASSESSMENT RUBRIC AT2 — Visualization Analysis Report

CO5 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO5-AT2 Visualization Analysis Report problems, in which students critique a described visualization design choice, identify its specific flaw, and justify a well-reasoned improvement.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the relevant visualization principle being evaluated (e.g., proportional ink, axis scale choice, overplotting, color scale design, accessibility).	Good understanding of the relevant principle, with only minor gaps.	Basic understanding, but with some misconceptions about which principle applies.	Poor understanding of core principles; misidentifies the nature of the flaw entirely.
Explanation	25%	Clearly explains why the design choice is flawed or effective, including its practical consequence for a reader — not just that it is ‘good’ or ‘bad.’	Explains the evaluation with adequate reasoning, covering most of the practical consequence.	Identifies the issue but explanation is generic or only loosely tied to its practical effect.	Little or no explanation provided; the evaluation is a bare assertion.
Application	20%	Proposes a specific, well-justified fix or recommendation that directly addresses the evaluated issue.	Proposes a workable fix, with only minor gaps in justification.	Proposes a fix, but it only partially addresses the issue or lacks justification.	Fails to propose a workable fix, or it does not address the evaluated issue.
Organization	15%	The response is clearly structured,	The response is organized, with	Basic structure; both parts are	Poorly organized; the response is

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		addressing critique and recommendation in a logical sequence with coherent overall flow.	only minor issues in flow or clarity.	addressed but the response lacks clarity or sequence.	incoherent, and one or both parts are left unaddressed.
Accuracy	10%	Both the critique and the recommendation are completely accurate and well-suited to the specific scenario described.	Mostly accurate, with only minor omissions or a small error.	Partially correct; the critique or recommendation contains notable inaccuracies.	Largely incorrect or inappropriate, with major gaps in both the critique and the recommendation.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 31 – Earthquake Energy and Logarithmic Axis

Introduction

Earthquake magnitudes can be associated with energy release values that vary over a very large numerical range. When values differ by several orders of magnitude, a normal linear axis can make the smaller values difficult to see. A **logarithmic axis** is therefore more suitable.

Given Data

Energy release values:

8, 250, 6000, 180000

Step 1: Calculate Consecutive Ratios

First ratio

$250/8=31.25$

So, the second value is **31.25 times** the first.

Second ratio

$6000/250=24$

So, the third value is **24 times** the second.

Third ratio

$180000/6000=30$

So, the fourth value is **30 times** the third.

Summary

Consecutive Values	Ratio
250 / 8	31.25
6000 / 250	24
180000 / 6000	30

Evidence

The values increase from:

$8 \rightarrow 250 \rightarrow 6000 \rightarrow 180000$

The overall range is:

$180000 = 22500$

Therefore, the largest value is **22,500 times** the smallest value.

Interpretation

On a linear scale, the value 180,000 would dominate the graph and the smaller values such as 8 and 250 would appear very close to zero.

A logarithmic axis is necessary because it represents multiplicative changes more effectively. Similar ratios can appear as similar distances on the axis.

For example:

$8 \rightarrow 250 \approx 31.25 \text{ times}$
 $250 \rightarrow 6000 = 24 \text{ times}$
 $6000 \rightarrow 180000 = 30 \text{ times}$

This reveals the structure of the data much more clearly than a linear scale.

R Code

```
R
energy <- c(8, 250, 6000, 180000)

ratios <- energy[-1] / energy[-length(energy)]

print(ratios)

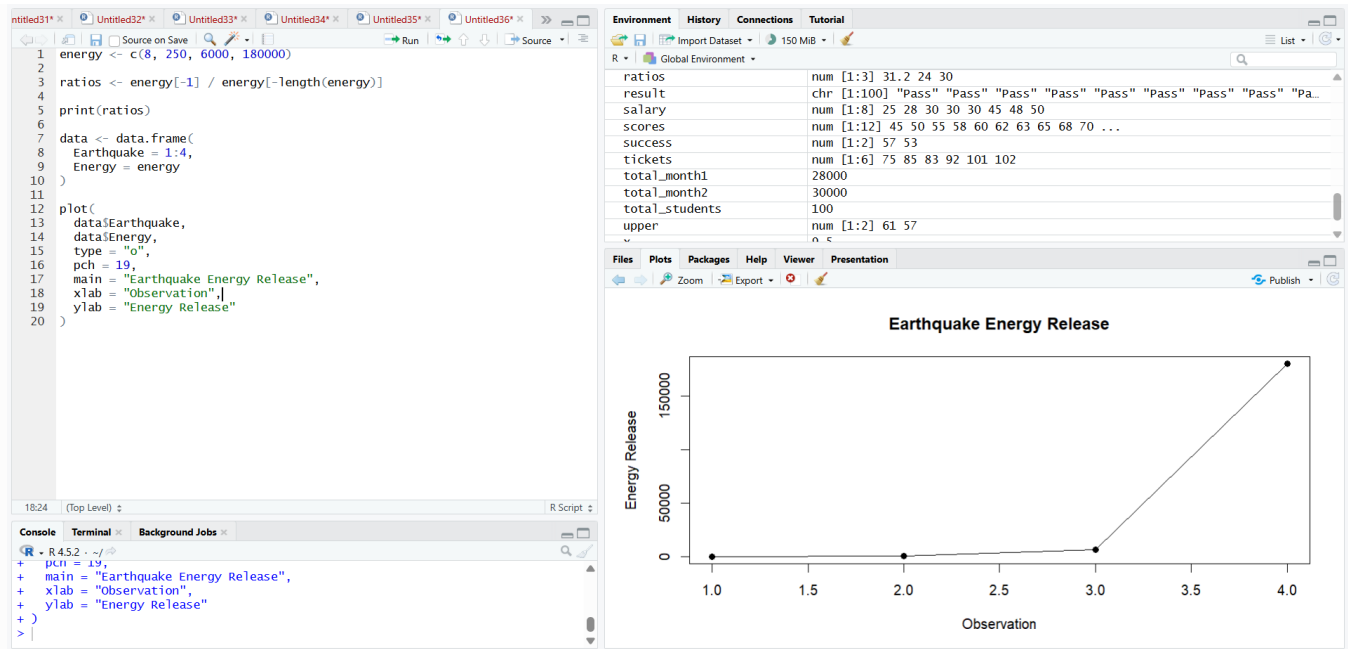
data <- data.frame(
  Earthquake = 1:4,
  Energy = energy
)

plot(
  data$Earthquake,
```

```

data$Energy,
type = "o",
pch = 19,
main = "Earthquake Energy Release",
xlab = "Observation",
ylab = "Energy Release"
)

```

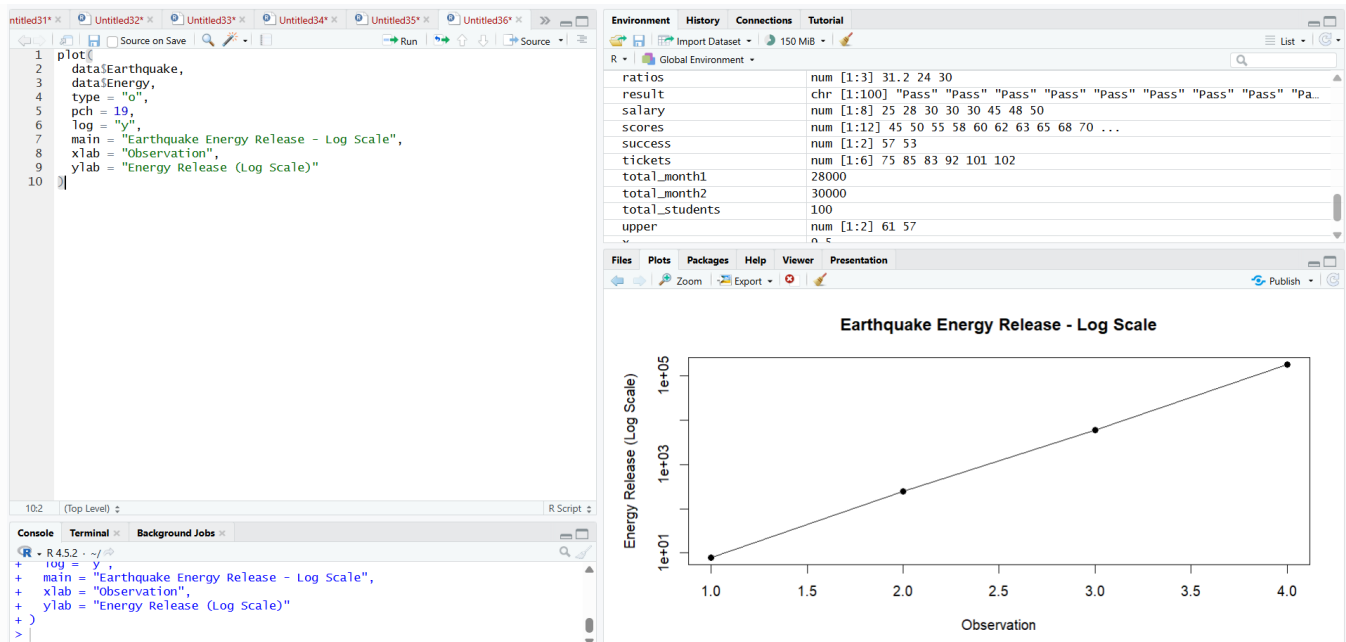


R Code with Logarithmic Axis

```

R
plot(
  data$Earthquake,
  data$Energy,
  type = "o",
  pch = 19,
  log = "y",
  main = "Earthquake Energy Release - Log Scale",
  xlab = "Observation",
  ylab = "Energy Release (Log Scale)"
)

```



Conclusion

The consecutive ratios are **31.25, 24, and 30**. Since the values span a very large range and increase by large multiplicative factors, a **logarithmic axis is necessary** to display all values clearly.

Problem 32 – 10×10 Waffle Chart

Introduction

A **waffle chart** represents percentages using a grid of equally sized cells. A 10×10 waffle chart contains:

$$10 \times 10 = 100$$

cells.

Therefore, **1 cell represents 1%**.

Given Data

Category	Percentage
Passed	76%
Failed	15%
Absent	9%
Total	100%

Cell Calculation

Passed

76%=76 cells

Failed

15%=15 cells

Absent

9%=9 cells

Therefore:

76+15+9=100

Exact Waffle Allocation

Category	Percentage	Number of Cells
Passed	76%	76 cells
Failed	15%	15 cells
Absent	9%	9 cells
Total	100%	100 cells

How to Build the 10×10 Chart

Create a grid containing **100 cells**.

Fill:

- First **76 cells** → Passed
- Next **15 cells** → Failed
- Last **9 cells** → Absent

R Code

```
R
library(ggplot2)

category <- c(
  rep("Passed", 76),
  rep("Failed", 15),
  rep("Absent", 9)
```

```

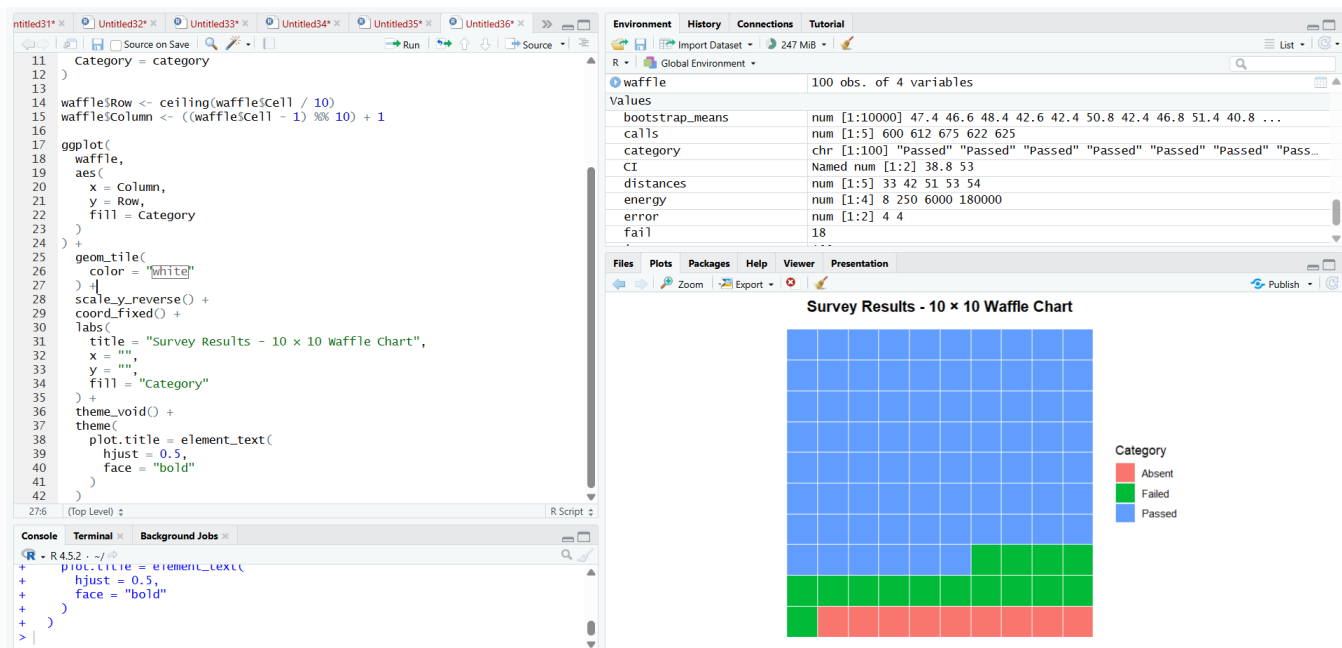
)

waffle <- data.frame(
  Cell = 1:100,
  Category = category
)

waffle$Row <- ceiling(waffle$Cell / 10)
waffle$Column <- ((waffle$Cell - 1) %% 10) + 1

ggplot(
  waffle,
  aes(
    x = Column,
    y = Row,
    fill = Category
  )
) +
  geom_tile(
    color = "white"
  ) +
  scale_y_reverse() +
  coord_fixed() +
  labs(
    title = "Survey Results - 10 × 10 Waffle Chart",
    x = "",
    y = "",
    fill = "Category"
  ) +
  theme_void() +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold"
    )
  )
)

```



Interpretation

The waffle chart allows the viewer to see that out of every 100 responses:

- **76 passed**
- **15 failed**
- **9 were absent**

The large majority of the cells represent the Passed category.

Conclusion

A 10×10 waffle chart contains **100 cells**, with each cell representing 1%. Therefore, **76 cells are assigned to Passed, 15 to Failed, and 9 to Absent.**

Problem 33 – 20,000 Stock Trades and Overplotting

Introduction

A scatterplot is useful for displaying the relationship between two numerical variables. However, plotting **20,000 stock trades** individually can result in severe overplotting, where many observations overlap and form a dense region or solid blob.

Evaluation

The raw scatterplot is not effective if the 20,000 observations overlap heavily.

The main problems are:

- Individual trades cannot be distinguished.
- High-density regions are hidden.
- Patterns and clusters become difficult to identify.
- The relationship between price and volume may not be visible.
- Outliers may be obscured.

Therefore, a different visualization technique should be used.

Technique 1 – Hexbin Plot

A **hexbin plot** divides the plotting area into hexagonal cells.

Each cell counts how many observations fall inside it.

The visual intensity of each cell represents the number of trades.

Advantage

It converts thousands of overlapping points into an easily interpretable **density representation**.

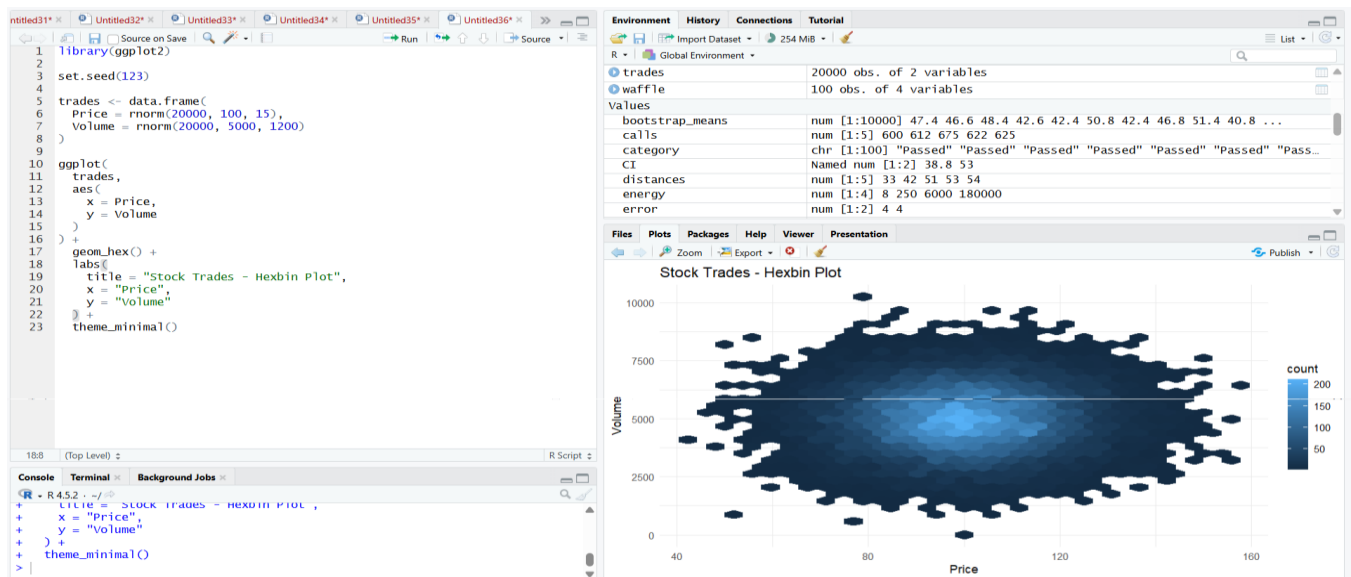
R Code – Hexbin

```
R
library(ggplot2)

set.seed(123)

trades <- data.frame(
  Price = rnorm(20000, 100, 15),
  Volume = rnorm(20000, 5000, 1200)
)

ggplot(
  trades,
  aes(
    x = Price,
    y = Volume
  )
) +
  geom_hex() +
  labs(
    title = "Stock Trades - Hexbin Plot",
    x = "Price",
    y = "Volume"
  ) +
  theme_minimal()
```



Technique 2 – Transparency

Another solution is to use **partial transparency**.

Instead of making every point completely opaque, reduce its opacity using `alpha`.

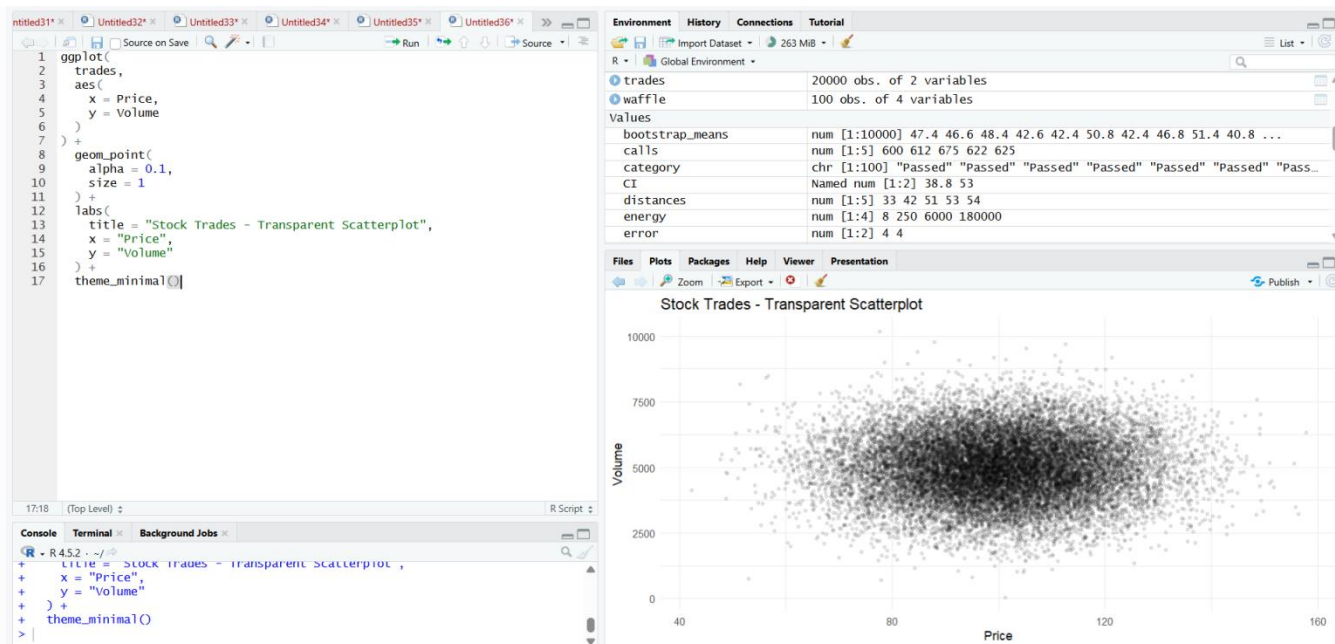
When many points overlap, the region becomes darker. Areas with fewer observations remain lighter.

Advantage

This allows the viewer to visually identify areas with greater concentration without completely hiding individual observations.

R Code – Transparency

```
R
ggplot(
  trades,
  aes(
    x = Price,
    y = Volume
  )
) +
  geom_point(
    alpha = 0.1,
    size = 1
  ) +
  labs(
    title = "Stock Trades - Transparent Scatterplot",
    x = "Price",
    y = "Volume"
  ) +
  theme_minimal()
```



Comparison

Technique	How It Works	Main Benefit
Hexbin	Groups observations into hexagonal cells	Clearly shows density
Transparency	Makes overlapping points partially transparent	Reveals concentration while retaining individual points

Conclusion

For 20,000 stock trades, a raw scatterplot can suffer from severe overplotting. Two suitable solutions are **hexbinning**, which summarizes observations into density cells, and **transparency**, which allows overlapping points to reveal dense regions through darker areas.

Problem 34 – Jittering + Transparency

Introduction

A categorical scatterplot can suffer from **point stacking** when many observations have the same categorical x-position. If all points are plotted at exactly the same position above each category, they overlap and hide the number and distribution of observations.

Evaluation

The chart has heavy stacking directly above each category label.

For example:



•
•
•
•
•
|
Branch A

A diagram illustrating heavy stacking. Five black dots are arranged vertically above the text 'Branch A'. A vertical line connects the bottom dot to the text label.

The viewer cannot easily determine how many observations are present because many points are drawn directly on top of each other.

Jittering

Jittering adds a small random horizontal displacement to the points.

Instead of:



•
•
•
•

A diagram showing four black dots arranged vertically, representing the original data without jittering.

the points become:



• •
• •
• •
•

A diagram showing the same four data points as the previous one, but with jittering applied. The first three points are now pairs of dots slightly offset horizontally, while the fourth point remains a single dot at the bottom.

The observations remain associated with the same category, but their positions are separated enough to become visible.

Partial Transparency

Transparency makes each point partially see-through.

When several points overlap, the area becomes darker or more visually concentrated.

Therefore:

- Few overlapping points → lighter appearance
- Many overlapping points → darker appearance

Why Combining Both Is Better

Using **jitter** + **transparency** provides two types of information:

Jitter

Shows the **individual observations**.

Transparency

Shows the **density of overlapping observations**.

Together, they allow the reader to understand both the distribution and concentration of ratings within each store branch.

R Code

```
R
library(ggplot2)

set.seed(123)

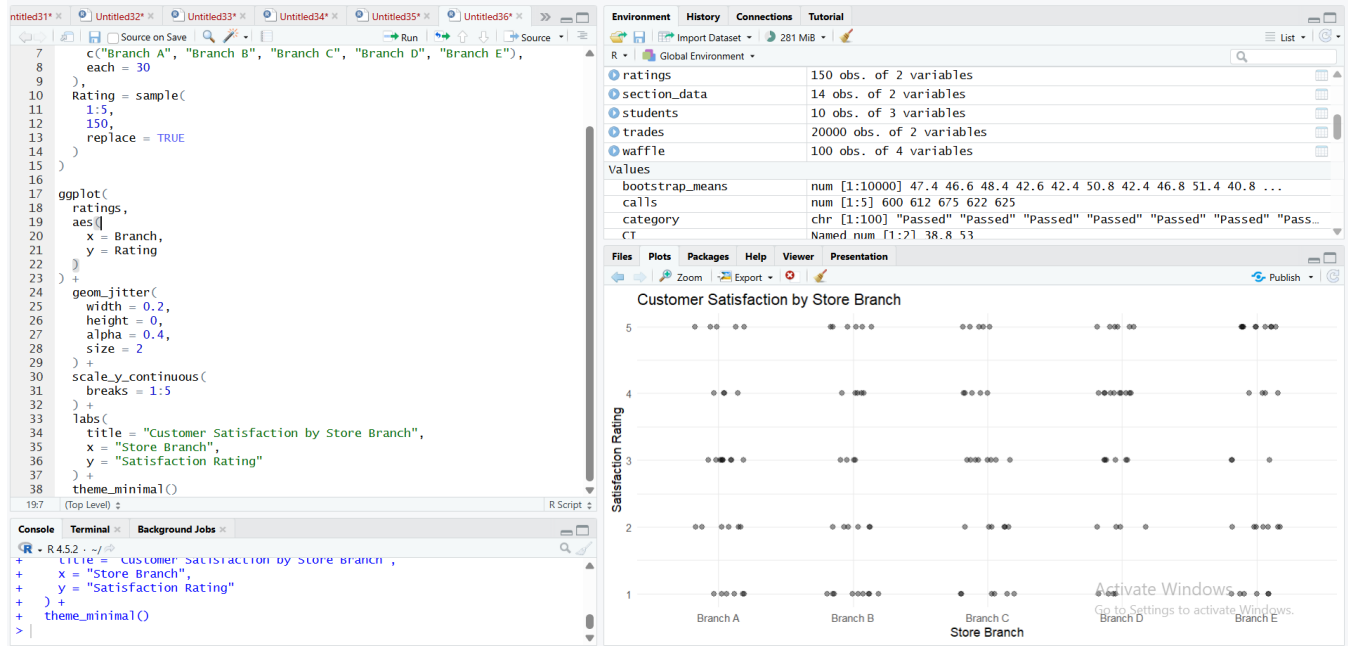
ratings <- data.frame(
  Branch = rep(
    c("Branch A", "Branch B", "Branch C", "Branch D", "Branch E"),
    each = 30
  ),
  Rating = sample(
    1:5,
    150,
    replace = TRUE
  )
)

ggplot(
  ratings,
  aes(
```

```

  x = Branch,
  y = Rating
)
) +
geom_jitter(
  width = 0.2,
  height = 0,
  alpha = 0.4,
  size = 2
) +
scale_y_continuous(
  breaks = 1:5
) +
labs(
  title = "Customer Satisfaction by Store Branch",
  x = "Store Branch",
  y = "Satisfaction Rating"
) +
theme_minimal()

```



Interpretation

Jittering separates points that would otherwise overlap exactly, making individual ratings visible.

Transparency additionally shows where observations are concentrated. A darker region indicates that many observations occupy a similar area.

Therefore, the combination improves readability without removing observations.

Conclusion

Combining **jittering with partial transparency** reduces the visual problem of point stacking. Jitter reveals individual observations, while transparency communicates density. Together they make categorical scatterplots easier to interpret.

Problem 35 – 2D Histogram

Introduction

A **2D histogram** is useful when a dataset contains a large number of pairs of numerical observations and a normal scatterplot becomes too dense.

Here there are **9,000 temperature and ice-cream-sales pairs**, so individual points may overlap heavily.

The 2D histogram solves this by dividing the graph into rectangular cells or bins.

How a 2D Histogram Works

The graph has:

- **X-axis → Temperature**
- **Y-axis → Ice Cream Sales**

The plotting area is divided into cells.

Each cell counts how many observations fall within its temperature and sales ranges.

What Does Cell Color Represent?

The **color of each cell represents the number or density of observations inside that cell.**

For example:

Light cell → Few observations

Medium cell → More observations

Dark cell → Many observations

Therefore, color is being used as a measure of **frequency/density**.

What Does a Very Dark Cell Mean?

A very dark cell indicates that **many of the 9,000 observations fall within that particular combination of temperature and ice-cream sales values.**

For example, suppose a dark cell corresponds approximately to:

Temperature = 30–32°C

Sales = 800–1000 units

This means many observations occurred when temperatures were between 30–32°C and sales were between 800–1000 units.

Interpretation

The 2D histogram allows the viewer to identify where observations are concentrated.

A series of dark cells moving upward from left to right could suggest a **positive relationship** between temperature and ice-cream sales.

However, the dark cell itself only indicates **high frequency/density**; it does not automatically prove causation.

R Code

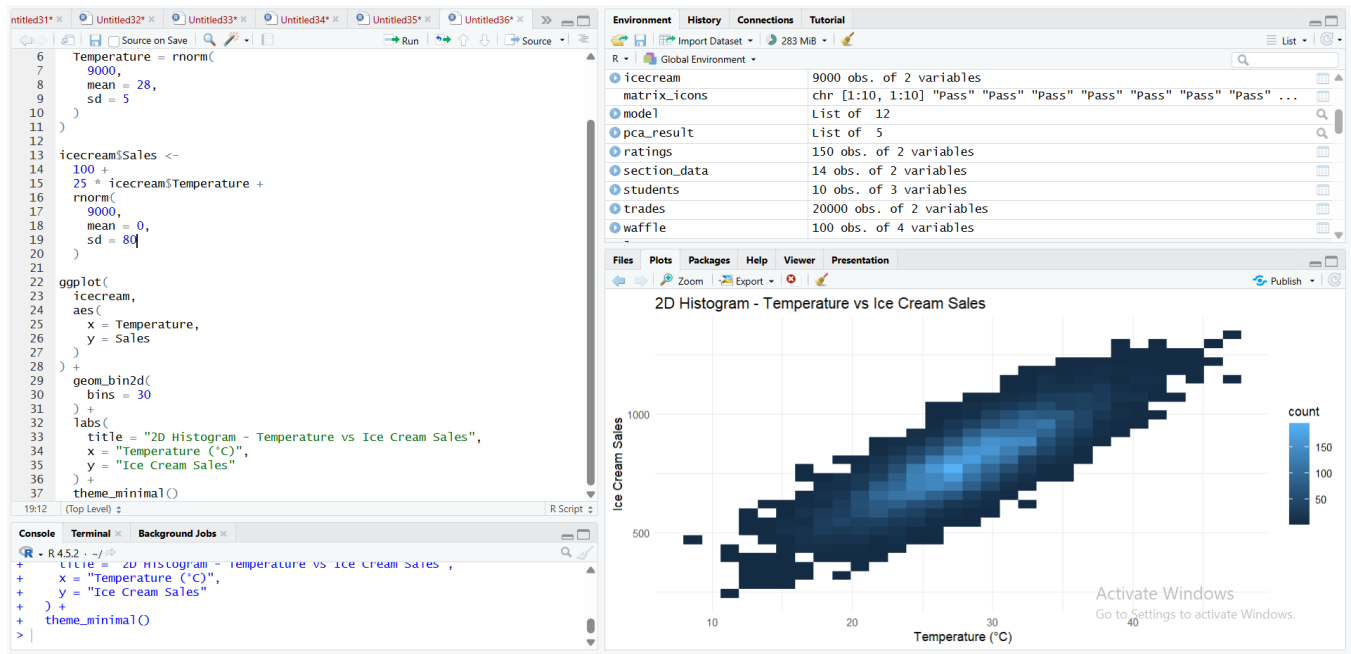
```
R
library(ggplot2)

set.seed(123)

icecream <- data.frame(
  Temperature = rnorm(
    9000,
    mean = 28,
    sd = 5
  )
)

icecream$Sales <-
  100 +
  25 * icecream$Temperature +
  rnorm(
    9000,
    mean = 0,
    sd = 80
  )

ggplot(
  icecream,
  aes(
    x = Temperature,
    y = Sales
  )
) +
  geom_bin2d(
    bins = 30
  ) +
  labs(
    title = "2D Histogram - Temperature vs Ice Cream Sales",
    x = "Temperature (°C)",
    y = "Ice Cream Sales"
  ) +
  theme_minimal()
```



Conclusion

In a 2D histogram, each cell's color represents the **number or density of observations** within that temperature-sales range. A **very dark cell indicates a high concentration of observations**, meaning many of the 9,000 data pairs fall within that particular region.